Title: Registered Report: A Large Experimental Test of the Effectiveness of Highly Cited Climate Change Messaging Strategies

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Abstract

Climate change presents a global threat to human well-being, making the identification of effective ways to communicate with the public about climate change a critical area for social science research. Results of five original pilot studies (N=3,216) we conducted, however, called into question the robustness of several widely-cited climate messaging strategies, possibly because the original findings were false positives or because the psychology of climate change attitudes has shifted. Further, research has not yet examined the relative effectiveness of these messaging strategies. To fill these gaps, we present a *registered report*–a paper with introduction and methods pre-registered and peer-reviewed before data collection–in which we experimentally tested (N=13,544) the effects of the ten highest cited climate change messaging strategies on Americans' beliefs, attitudes, and behavioral intentions related to climate change. We found the climate messages had many significant effects on beliefs, attitudes, and behavioral intentions, though persuasive effects were generally small, similar in size for different messages, and messages did not increase pro-environmental donations. Notably, persuasiveness of messages varied little across party or ideological lines, inconsistent with the theoretical reasoning underpinning several messaging strategies. Exploratory analyses suggest the messages treated several possible mechanisms, limiting theoretical inference. Taken together, our study identifies multiple messaging strategies that were consistently persuasive across a range of pro-environmental attitudes, though the magnitude of effects highlights the limits of relying on persuasion of the mass public alone to drive action on climate change in the U.S.

Introduction

Although environmental experts are highly concerned that some of the most serious consequences of climate change – more-extreme weather events and sea level rise – will soon become irreversible (Intergovernmental Panel on Climate Change, 2022), many people around the world do not perceive climate change to be a major threat (Fagan & Huang, 2019). In the United States, the country with the most CO_2 emissions in the world (Union of Concerned Scientists, 2022), many scholars believe that - given the outsized influence of fossil fuel companies on politicians via campaign donations - legislation addressing climate change in the US will only be achieved via a large majority of Americans demanding action (Brulle & Downie, 2022). However, only 25% of Americans think that global warming is extremely personally important (Krosnick & MacInnis, 2020), only 34% of likely voters believe that limiting carbon dioxide emissions should be a higher priority for the President and Congress than increasing oil and gas drilling (Rasmussen Reports, 2022), and even Democrats who widely agree that the U.S. needs a big environmental program would rather compromise on the environment than other culture war issues (Vavreck et al., 2019).

Accordingly, the question of how to effectively communicate with the public about climate change has been analyzed and debated for decades by academics, as well as policymakers and analysts beyond the academy, commanding a large body of work on effective messaging approaches (for reviews, see Rode et al., 2021 and Wong-Parodi & Feygina, 2020). The results of such research is important because it shapes how journalists with large audiences write about climate change (e.g., Wallace-Wells, 2017). This body of work has resulted in some optimism that policy makers, journalists, and activists can convince the public to prioritize mitigating climate change.

However, there is reason to think some of the most cited messaging strategies in this line of research may not, or no longer, have robust effects. We conducted five (total *N*=3,216) pilot replication studies – detailed below in the Pilot Data section – which showed null effects of previously published messaging strategies (Feinberg & Willer, 2011, 2013; Wolsko et al., 2016), suggesting that these widely-cited messaging strategies may no longer be effective, at least in their previously published form.

There are many potential explanations for the failure of previously published messaging strategies to replicate. One potential explanation is that the temporal validity of climate change messaging strategies is low. Temporal validity is the extent to which a result can be generalized across different times (Munger, 2019). Multiple developments may have reduced the effectiveness of climate change messaging strategies over time, including the increased polarization of public opinion on the issue (e.g., Dunlap, McCright, & Yarosh, 2016) or the increased saturation of the information environment. Additionally, the politicization and polarization of climate change in the news may have caused the psychological underpinnings of climate change attitudes to change (Chinn, Hart, & Soroka, 2020). An alternative possibility is that the original studies were underpowered, yielding false positive results, and – as a result – are unreliable. Yet another possibility is that the replication studies were underpowered, yielding false negative results.

Another obstacle to effective communication on climate change is that the relative effectiveness of these messaging strategies is unknown. Papers typically focus on testing single messaging strategies, using their own dependent variables and sampling populations. A recent meta-analysis did not find evidence that the type of messaging strategy influenced effect sizes (Rode et al., 2021), while a systematic review identified psychological distance, elite cues, and ideological fit as the three most important ingredients for effective messaging strategies (Rode & Ditto, 2021). However, without clean experimental designs that hold measurement and sampling constant, it is impossible to compare the effect sizes of published messaging strategies in a highly controlled way. As a result, it remains a major limitation of the literature that we cannot infer the *most* promising messaging strategies in the field.

A third limitation of prior research is that it has typically focused on increasing belief in climate change without assessing whether effective messaging strategies also increase support for climate change mitigation policies or pro-environmental behaviors (see Tables S3-S7). To more thoroughly assess the effects of messaging strategies, we measure belief in climate change, climate change concern, support for general climate change mitigation policies, and intention to engage in political pro-environmental behaviors. We define (a) *belief in climate change* as the beliefs that the world is warming and that this warming is human-caused; (b) *climate change concern* as the perceived seriousness of climate change; (c) s*upport for general climate change mitigation policies* as general support for policies intended to reduce climate change or its consequences; and (d) *political pro-environmental behavioral intentions* as the intention to personally engage in political behaviors intended to reduce climate change or its consequences. We selected these outcomes of interest because they include the most commonly targeted by climate change messaging strategies.

In this paper, we seek to answer the question: can highly-cited climate change messaging strategies be leveraged to increase Americans' belief in, concern about, support for policies to address, and/or intentions to engage in political behaviors to mitigate climate change? To this end, we assessed both the robustness and relative effectiveness of ten highly cited messaging strategies in the published literature on climate change communication. We used an experimental approach with a commensurate design, testing the messaging strategies under identical conditions, including using the same measures of outcome variables, and recruiting a national, non-probability sample of participants that is representative of US population benchmarks for a number of key demographic variables (e.g., gender, age, race/ethnicity, education, region, and party identification).

From 157 papers that we identified in the literature on climate change messaging, we selected ten highly cited messaging strategies that found statistically significant changes on at least one of our outcomes of interest (see *Methods* section for details on the selection process). The selected messaging strategies (see the *Supporting Information* (SI) for details) represent a variety of approaches to climate change communication, including some of the most frequently studied types of message frames (Badullovich, Grant, & Colvin, 2020). Specifically, five of the selected messaging strategies sought to change people's attitudes by describing supporting climate change mitigation as being consistent with their existing moral values and ideologies, such as care and considerateness (Bain, Hornsey, Bongiorno, & Jeffries, 2012), free market beliefs (Campbell & Kay, 2014), belief in a just world (Feinberg & Willer, 2011), patriotism (Wolsko et al., 2016), purity (Feinberg & Willer, 2013), and system preservation (Feygina, Jost, & Goldsmith, 2010). Two of the selected messaging strategies emphasized that climate change is scientific consensus (Cook, Lewandowsky, & Ecker, 2017; van der Linden et al., 2017). Two other messaging strategies focused on the costs (Hart & Nisbet, 2012) or gains (Spence & Pigeon, 2010) associated with climate change mitigation for participants' communities.

Replication studies need to balance the tradeoff between maximizing the chances for treatment effects and replicating the original studies as closely as possible. Using the original materials is an important principle for replication studies (hence the term "replication").

However, exact replications using original but outdated materials may undermine treatment effects and preclude our chances of identifying currently relevant and effective solutions to improving climate change opinion. An alternative is the "conceptual" replication, which attempts to test the theoretical basis of a stimuli by keeping it close to the original study but updating it in a common sense manner to conform to the contemporary environment. Thus, adapting outdated references in the original materials may restore the viability of the messaging strategies in its temporal context and thus get us closer to a replication. Adapting the original materials also allowed us to remove any deception from the treatments, an important limitation of some of the original materials. To address this, we collaborated with the developers of the original materials, offering them the opportunity to revise their materials[1] to fit the present moment, while maintaining the theoretical basis of the messaging strategy. We carefully assessed these revisions to ensure that the targeted manipulated theoretical construct remains unaltered.

Research Questions & Hypotheses

Table 1 shows our research questions, hypotheses, and analysis plans. Our study assessed the effectiveness of the ten climate change messaging strategies in several ways. To test the absolute effectiveness of the messaging strategies, we compared participants' levels on the outcomes in each treatment condition to participants in a control condition across the full sample. For each messaging strategy, we tested the hypothesis that the corresponding treatment significantly increased (a) belief in climate change, (b) climate change concern, (c) support for general climate change mitigation policies, and (d) political pro-environmental behavioral intentions among Americans.

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[1] To acknowledge their work in developing the treatments, we offered authorship to all authors of the original materials.

We also tested the absolute effectiveness of the messaging strategy for Democrats and Republicans, separately. Of the ten original papers, eight found their respective treatment effects depended on people's political beliefs, with stronger effects among participants with a Republican partisan identity (Campbell & Kay, 2014; Hart & Nisbet, 2012; van der Linden et al., 2017), a conservative political ideology (Feinberg & Willer, 2013; Wolsko et al., 2016), strong belief in a just world (Feinberg & Willer, 2011), high levels of system justification (Feygina et al., 2010), and strong free market support (Cook et al., 2017). Measuring these many moderators would repeatedly prime participants with ideological beliefs before they read the treatment materials, potentially impacting the effects of the treatment. Here, we focus on partisanship because it is the most common moderator among the ten original papers, is generally correlated with the other beliefs, and is information that is more often available to practitioners intervening in the real world than the other beliefs. Because of the difficulty of sufficiently powering tests of interaction effects (Sommet, Weissman, Cheutin, & Elliot, 2022), we investigated the possibility of these patterns by testing for the presence or absence of significant persuasive treatment effects among Republicans, for which we are adequately powered (see *Methods* section).

To test the absolute effectiveness of the messaging strategies among Republicans, we compared Republicans' levels on the outcomes in each treatment condition to Republicans in the control condition. To test the absolute effectiveness of the messaging strategies among Democrats, we compared Democrats' levels on the outcomes in each treatment condition to Democrats in the control condition. For each treatment, we tested the hypotheses that this messaging strategy significantly increases (a) belief in climate change, (b) climate change concern, (c) support for general climate change mitigation policies, and (d) political pro-environmental behavioral intentions among (i) Republicans and (ii) Democrats.

To test the relative effectiveness of the messaging strategies, we compared participants' levels on the outcomes in each treatment condition to participants in the other treatment conditions. We did these tests for the full sample and among Democrats and Republicans separately. Our (exploratory) research questions for assessing the relative effectiveness are: Which messaging strategies *most strongly* increase (a) belief in climate change, (b) climate change concern, (c) support for general climate change mitigation policies, and (d) political pro-environmental behavioral intentions among (i) Americans, (ii) Democrats, and (iii) Republicans?

Methods

Ethics information

The study has been approved by the Institutional Review Board at Stanford University. All participants provided informed consent. Participants were paid a small monetary fee for their participation. Neither the experimental treatments nor the measures in this study used deception. Because the study was conducted online, there was no interaction between the experimenter and participants.

Replication Approach

Our replication approach follows two principles that shaped our methodological decisions. The first principle is generosity towards the original findings. Thus, rather than focusing on direct replication of prior findings using prior materials, we are interested in identifying messaging strategies that work in the contemporary U.S. Therefore, to give the messaging strategies the best chance to successfully improve climate change attitudes, we allowed authors to revise their original materials to update outdated materials because several studies (e.g., Feinberg & Willer, 2011 and Campbell & Kay, 2014) featured materials with concrete references that are now outdated, or which are region-specific and do not apply to a national-level sample (e.g., Hart & Nisbet, 2012; Spence & Pidgeon, 2010). Note, the primary authors supervised authors' revision of study materials to ensure the materials tested maintained the underlying theoretical rationale of the messaging strategy. Additionally, we included a "Points of Dissent" section in the SI where prior authors express differences of opinion regarding methodological choices, statistical analyses, inferences from results, or any other aspect of the paper. The second principle is comparability. That is, we are interested in assessing not only the presence or absence of effects, but also relative treatment effects in order to identify the *most* successful messaging strategies. Therefore, we used consistent dependent variables across all treatment conditions instead of the original dependent variables.

Messaging Strategies

The messaging strategies that were tested in our study were determined using four steps. First, we sought to identify all published experiments on climate change messaging. We only sought empirical papers with experiments on human subjects that collected at least one of the following four variables: (a) belief in climate change, (b) climate change concern, (c) support for general climate change mitigation policies, and (d) political pro-environmental behavioral intentions. We identified 76 papers from a recently published meta-analysis (Rode et al., 2021) and 38 additional papers from a recently published review article (Rode & Ditto, 2021) on interventions affecting climate change attitudes. We identified 18 additional papers via advice from two experts in the field. Finally, we identified 25 additional papers by searching journals (*PNAS* and *Nature Climate Change*) known to publish high impact studies on climate change messaging with the key words "climate change," "belief," "concern," and "deny." Following these steps, we identified 157 papers. The full list of identified papers is available at https://osf.io/a5bup.

Second, we collected citation counts for each of these 157 papers. We calculated three metrics that varied in the weight assigned to older citations. The first metric is the total number of citations on Google Scholar as of December 31, 2021. The second metric is the number of citations on Google Scholar in the preceding five years (from January 1, 2017 to December 31, 2021). The third metric is the number of citations per year since publication. We found that the rank scores of the three metrics were highly correlated (all rs > .91). Seven treatments would have been selected independent of the metric used. In other words, different selection strategies that put various penalties on older citations resulted in similar selected treatments.

Third, using the average rank across the three metrics, we selected the top ten messaging strategies that found statistically significant changes on at least one of our outcomes of interest. We selected ten because that was within the limits of our budget. When there was a large conceptual overlap between messaging strategies from two or more papers, we only included the treatment from the most cited paper.

Fourth, the authors of the original papers were offered the opportunity to revise their treatments. Nine of ten teams did in fact revise their treatments. This step allowed the authors to adjust their treatments to the new sampling population, the specific dependent variables, and the current environment. For example, some treatments contained content that focused on countries other than the US. Some treatments were tested more than 10 years ago, and referenced material that is outdated (e.g., climate change reports that have since been updated). Some treatments used deception. The primary authors of this paper and the reviewers assessed the revisions of the treatment to ensure that the originally manipulated theoretical construct remains unchanged. The revised and original treatments are available in the SI.

Procedure & Design

The experiment has a 11 (condition; between-subjects) x 2 (pre vs post; within-subjects) design. The procedure of the experiment contained the following steps.

First, we measured our main dependent variables of interest. The order of the main dependent variables was randomized. For each main dependent variable, the order of the items was randomized. Belief in climate change was measured with three items (e.g., “Do you think that the world’s temperature probably has been going up over the past 100 years, or do you think this probably has not been happening?”). Climate change concern was measured with three items (e.g., “How serious a problem is climate change?”). Support for general climate change mitigation policies was measured with four items (e.g., “The U.S. government should do more to reduce greenhouse gas emissions”). Political pro-environmental behavioral intentions was measured with four items asking about the likelihood to engage in pro-environmental activities in the next twelve months (e.g., “sign a petition in support of protecting the environment”).

Second, we collected secondary and exploratory dependent variables. The goal of collecting these variables is to test whether potential effects of the treatments are robust to practical tradeoffs. The order of the secondary dependent variables was randomized. For each secondary dependent variable, the order of the items was randomized. Support for specific climate change mitigation policies was measured with four items featuring climate mitigation policies, that also highlight likely consequences (e.g., “Ban the construction of new coal-fired power plants. (This policy would decrease greenhouse gas emissions but might increase energy costs for the average household.)”). Support for pro-environmental candidates was measured with four items in which participants need to choose between a pro-environmental candidate who holds the opposing view from the participant on another issue and an anti-environmental candidate who holds the same view as the participant on another issue. Support for company-led climate change mitigation was measured with three items featuring climate change mitigation based on voluntary leadership of companies instead of the government (e.g., “Companies should do more to reduce greenhouse gas emissions”). Non-political pro-environmental behavioral intentions was measured with six items featuring climate change mitigation based on individual actions in non-political contexts (e.g., “Eat less meat”; adapted from Brick et al., 2017). Donations to pro-environmental organizations were measured with a single item, asking participants how many of 100 cents they wanted to donate to one of five leading pro-environmental organizations. We used the sum of the donations to the five organizations as our measure of donations to pro-environmental organizations. All items were measured on 101-point slider scales. Such scales are more sensitive for detecting nuanced differences (Evangelidis, 2023) and previous research suggests that 101-point scales allow detecting small treatment effects (Voelkel et al., 2023). The full list of items is available in Tables S2-S5.

Third, participants completed a short demographic questionnaire, including a measure of their partisanship. Partisanship was measured with the standard questions from the American National Election Studies. This measure distinguishes eight categories: strong Democrats, not very strong Democrats, Independents closer to the Democratic Party, Independents closer to the Republican Party, not very strong Republicans, strong Republicans, those identifying with neither party, and those identifying with other parties). We pooled these eight categories into three categories: Democrats (including strong Democrats, not very strong Democrats, and Independents closer to the Democratic Party), Republicans (including strong Republicans, not very strong Republicans, and Independents closer to the Republican Party), and Others (those identifying with neither Democrats nor Republicans).

Fourth, participants were randomly assigned to either one of the treatment conditions or the control condition. In each of the ten treatment conditions, participants were asked to read the corresponding treatment materials. In the control condition, participants were randomly assigned to read one of three neutral messages that are unrelated to climate change. The control conditions were on the issues of (a) the history of neckties, (b) the rules of baseball, and (c) different types of dances. Using several different control conditions minimizes the chance that any observed effects are due to unexpected effects of a particular control condition. The length and design of the control conditions was matched to the length and design of the experimental treatments. The full materials for all conditions are available in the SI.

Afterwards, we measured our main and secondary dependent variables for a second time. We measured the main dependent variables first (in randomized order). Then we measured the secondary dependent variables (in randomized order). Finally, we collected responses to manipulation check items (one for each treatment condition; see Table S4 for the specific items) and other exploratory measures at the end of the survey.

Sampling Plan

Participants were recruited from a national, non-probability panel of American residents provided by Bovitz-Forthright. Bovitz-Forthright has provided samples for several similar published survey experiments that have found significant treatment effects (e.g., Corbett et al., 2022; Voelkel et al., 2023a; see Forthright, n.d., for a full list), including a similar many-treatment experiment (Voelkel et al, 2023b). The sample was quota-matched to the U.S.

population for several demographic variables, including gender, age, race/ethnicity, education, region, and party identification.

Data collection was stopped as soon as we had collected 13,000 participants. Participants were randomly assigned to one of eleven experimental conditions (10 treatment conditions, 1 control condition). Because we are particularly interested in the 10 comparisons of the treatment conditions to the control condition, we assigned more participants to the control condition. Following the rule developed by Dunnett (1955) to maximize power for the comparisons between participants in the control condition and the treatment conditions, we assigned approximately 1,000 participants to each treatment condition and approximately 3,000 participants to the control condition.

Power analyses using 10,000 Monte Carlo simulations suggest that this design with 1,000 participants in each treatment condition and 3,000 participants in the control condition provide us with more than 99% power (99% for subgroup analyses among Democrats; 98% for subgroup analyses among Republicans) to detect even small effect sizes of Cohen's $d \geq 0.10$ for treatment vs control comparisons in the full sample. For treatment vs treatment comparisons, we have more than 99% power (87% for subgroup analyses among Democrats; 88% for subgroup analyses among Republicans) to detect effect sizes of Cohen's $d \geq 0.10$ in the full sample. The effect sizes found in the original studies are typically larger than these benchmarks (see Table S7). The power analyses are not affected by the exclusion criteria described below because we recruited participants until we had collected 13,000 participants who completed the survey.

Analysis Plan

Exclusions

Participants were excluded for several reasons. First, we excluded participants who indicated that they are less than 18 years old because our IRB protocol prohibits surveying minors. Second, we excluded cases with duplicated participant ID (we only kept the first case) because taking the experiment multiple times prohibits participants' blindness to the study design. Third, we excluded participants who failed attention checks to increase the rates of participants who pay attention to the treatment and outcome variables. The attention checks were asked before treatment assignment to avoid post-treatment bias (Montgomery et al., 2018). Finally, out of necessity, we excluded participants who dropped out of the survey before treatment assignment.

Reliability of measures

We formed composites by averaging the items measuring each of our dependent variables of interest. We report Cronbach's alpha as our measure of reliability for all scales. Cronbach's alpha was above .70 for all measures (see Table S13)..

Missing data

We excluded participants who attrited after treatment assignment. A participant who had missing values for one or multiple primary dependent variables was still included in the analyses on the primary dependent variable(s) for which they provided data. We tested for differential attrition by condition in two ways (Lin & Green, 2016). First, we tested whether experimental condition affected attrition using a heteroskedasticity-robust F-test (Wooldridge, 2010). Second, we tested whether experimental condition moderated the effect of the pre-treatment measures of the outcomes on attrition using a heteroskedasticity-robust F-test. Because there was evidence for differential attrition, we conducted a robustness check by rerunning our preregistered models using inverse-probability weighting to correct for differential attrition. Inverse-probability weighting describes a procedure in which weights are generated that individuals who did not attrit from the study but had similar attributes as attriters are weighted more heavily to correct for the missing data from participants who did attrit. For each participant, we calculated the estimated probability of providing an outcome using random forests based on experimental condition and pre-treatment outcomes. We then calculated the weights as the inverse of these probabilities. Finally, we used these weights in our regression models.

Outliers

All values on our response scales are theoretically reasonable responses. Therefore, we did not exclude any potential outliers.

Absolute effectiveness: treatments versus control (preregistered)

We tested the effects of each of the treatments relative to the control condition with ordinary least squares regression with robust standard errors. We separately regressed the post-treatment measure of a dependent variable on the experimental condition (a series of dummy variables with the control condition serving as the reference category). We controlled for the pre-treatment measure of the dependent variable.

Absolute effectiveness among Democrats (preregistered)

We used the same analysis strategy as described in the section on absolute effectiveness. However, we only included participants who identify as Democrats.

Absolute effectiveness among Republicans (preregistered)

We used the same analysis strategy as described in the section on absolute effectiveness. However, we only included participants who identify as Republicans.

Relative effectiveness: treatments versus each other (preregistered)

We tested the relative effects of the treatments using the same regression model as for testing the effects of the treatments against the control but we switched out the reference category for the experimental condition variable. For each of the primary dependent variables, we first ordered the treatments by their effect sizes. We then tested the treatment with the strongest effect against the treatment with the second strongest effect (the 1-2 test). If this 1-2 test is significant, we tested the treatment with the second strongest effect against the treatment with the third strongest effect (the 2-3 test). If the 1-2 test is not significant, we tested the effect of the treatment with the strongest effect against the treatment with the third strongest effect (the 1-3). We followed this strategy until we reach a treatment that did not significantly differ from the control condition. This way we identified which treatments have stronger effects than other treatments using the fewest pairwise tests possible. We ran these analyses among the full sample, only including Democrats, and only including Republicans.

Inference criteria

We used p-values as our main inference criterion. We reported $p \leq .05$ as significant effects, $.05 < p \leq .1$ as marginally significant effects, and $p > .1$ as nonsignificant effects. We used two-tailed tests for all analyses. Although corrections for multiple tests are necessary when the same hypothesis is tested in multiple ways (e.g., if we were interested in the hypothesis that “at least one of the treatments reduces belief in climate change”), we do not correct for multiple comparisons because our interest is in the individual effects of the treatments on the different dependent variables.

We used the Bayes Factor as our secondary inference criterion to distinguish evidence favoring the null hypothesis from evidence favoring neither the null hypothesis nor the alternative hypothesis. We conducted Bayesian ANCOVAs using the same dependent variable, independent variable, and control variable as for the frequentist regression models. We conducted post-hoc tests for the experimental condition variable, using JASP's default priors without corrections for multiple hypothesis testing (i.e., setting the prior odds for null hypothesis versus alternative hypothesis to 1). We reported a Bayes factor between 1 and 3 as weak evidence, a Bayes factor between 3 and 10 as moderate evidence, and a Bayes factor of more than 10 as strong evidence (van Doorn et al., 2021).

Data Availability

The anonymized data for our study is publicly available via https://osf.io/va27p/.

Code Availability

The analysis code for our studies is publicly available via https://osf.io/va27p/.

Pilot Data

We conducted five pilot studies attempting to replicate three of the ten selected messaging strategies. In all three instances, the replication sample was at least 2.5 times the size of the original sample, in line with recommended practices for replication studies (Simonsohn, 2015). All five studies failed to replicate the original findings. Original studies failed to replicate, both when using identical, or near identical, materials to the original, or when using revised materials. These replication failures led us to believe that a more extensive replication project – the study we propose here – would be a valuable contribution to this very important literature. Below we summarize the replication studies and their findings. Detailed results can be found in the SI.

Replication of Feinberg & Willer (2013). Study 3 of the original paper (n = 308) reported that moral reframing of pro-environmental messages can improve climate change attitudes among conservatives but not among liberals. Specifically, conservative participants who were assigned to read a pro-environmental message rooted in values of moral purity and sanctity, relative to conservative participants who were assigned to read a pro-environmental message rooted in values of reducing harm and care and relative to conservative participants who were assigned to read a neutral message, reported higher belief in global warming as well as more pro-environmental attitudes, and support for pro-environmental legislation. Liberals were not significantly affected by the treatments. This influential work has been cited over 700 times to date.

The results of a replication study with 784 participants using original study materials did not replicate the original findings. Specifically, we did not find an interaction of political ideology and experimental condition (purity/sanctity vs. harm/care vs neutral) on belief in global warming, pro-environmental attitudes, and support for pro-environmental legislation. The main effects of the experimental condition were non-significant as well. In sum, we did not find evidence that the climate change messaging strategy of emphasizing moral purity improved climate change attitudes or behavioral intentions among American conservatives.

Replication of Feinberg & Willer (2011). Study 1 of the original paper ($n = 97$) showed that a dire messaging strategy on the disastrous impacts of climate change decreased belief in climate change among people who believe in a just world. However, the same dire messaging strategy increased belief in climate change among people who believe in a just world when it was combined with a positive ending that climate change was still solvable. Among people who do not believe in a just world, both messaging strategies increased belief in climate change similarly. This work has been cited over 500 times to date.

The three replication studies we conducted (total $n = 2,170$) were designed to be an improvement of the original study. In each study, participants were assigned to conditions that varied in the extent to which they described climate change as (a) a dire problem and (b) a solvable problem. This design allowed to disentangle the effects of each of the components of the original dire but solvable messaging strategy. The reasoning from the original study would yield a prediction for a three-way interaction effect (belief in a just world x problem condition x solution condition).

The results of the three replication studies did not replicate the original findings. The predicted three-way interaction effect was essentially zero and non-significant. The main effects and two-way interaction effects involving the two factors were also insignificant. The only exception was the main effect of the problem condition which suggested that participants in the more dire problem condition significantly increased belief in climate change relative to the less dire problem condition. In sum, we did not find evidence that the effect of the climate change messaging strategy emphasizing how dire or solvable climate change is on belief in climate change depends on people's belief in a just world.

Replication of Wolsko, Ariceaga, and Seiden (2016). Study 3 of the original paper (n = 97) reported that moral reframing of pro-environmental messages can improve climate change attitudes among conservatives but not among liberals. Specifically, conservative participants who were assigned to read a pro-environmental message rooted in binding moral values, relative to conservative participants who were assigned to read a pro-environmental message rooted in individualizing moral values, reported higher conservation intentions, improved climate change attitude measure, and stronger intentions to donate to a pro-environment group. Liberal participants in the individualizing condition reported higher conservation intentions and more improved climate change attitudes than liberal participants in the binding condition. There was no difference in intentions to donate to a pro-environment group among liberal participants. This work has been cited over 350 times to date.

The results of a replication study with 262 participants using original study materials did not replicate the original findings. Specifically, we did not find evidence for a two-way interaction effect (political ideology x experimental condition) on conservation intentions, climate change attitudes, and intentions to donate to a pro-environment group. The main effects of the experimental condition were non-significant as well. In sum, we did not find evidence that the climate change messaging strategy of emphasizing binding or individualizing values differentially impacted climate change attitudes or behaviors.

Results

Sample Size

24,977 participants started the survey. Following preregistered exclusion criteria to ensure attentive respondents, we excluded 11,156 participants before the treatment assignment. The remaining 13,821 participants were randomly assigned to a condition, of which 13,544 completed at least one of the primary outcomes.[2] The final sample was recruited to be representative of the population of adult residents of the U.S. with respect to a series of quotas

[2] Consistent with preregistered procedures, to minimize attrition and maximize power, we analyze responses from all participants who completed the relevant measures needed to estimate our models. Final sample sizes for all models are given in Table S11. Only 2% of the participants who were assigned to a condition attrited from the study without completing any of the primary outcomes, ranging from 0.9% to 4.4% across the different conditions (see Table S36). Using Wald tests with heteroscedasticity-consistent standard errors, we find statistically significant differences in attrition rates across conditions for most outcomes (see Table S37). Robustness checks using inverse-probability weighting find similar results as those reported in the main text (see Tables S38-S46). To further address potential threats to valid causal inference posed by differential attrition across conditions, we implemented a method (Voelkel et al., 2024) for collecting data from participants who attrited from the study after random assignment to conditions, re-recruiting them to a short study including only the key dependent variables. Robustness checks including recaptured participants and using imputation of pre-treatment-levels for attrited participants who were not recaptured, yield results similar to those reported in the main text (see Tables S47-S55).

for gender, age, race and ethnicity, education, region, and partisan identity categories (see Table S12).

Baseline Statistics

Figure 1 shows levels of belief in climate change, climate change concern, support for general climate change mitigation policies, and political pro-environmental behavioral intentions, as well as the five other outcomes, among participants in the control condition (N = 3,132; reliability estimates are available in Table S13). On average, participants reported moderate levels of pro-environmental attitudes, behavioral intentions, and behaviors, with average levels ranging from 32.5 to 70.9 (scale: 0-100): support for pro-environmental candidates (M = 32.5, SD = 21.5), political pro-environmental behavioral intentions (M = 33.9, SD = 28.9), support for specific climate change mitigation policies (M = 53.3, SD = 24.0), non-political pro-environmental behavioral intentions (M = 54.5, SD = 24.3), climate change concern (M = 60.4, SD = 31.7), pro-environmental donations (M = 61.6, SD = 45.3), belief in climate change (M = 65.4, SD = 22.5), support for general climate change mitigation policies (M = 68.0, SD = 29.3) and support for company-led climate change mitigation (M = 70.9, SD = 28.1). Descriptives for the other experimental conditions are reported in Table S16-S17. Democrats reported significantly higher levels on all outcomes than Republicans. Participants who identified with neither the Democratic nor the Republican Party reported significantly higher levels than Republicans and significantly lower levels than Democrats on all outcomes.

Confirmatory Analyses

Figure 2 visualizes the treatment effects on the primary outcomes for the entire sample, and also specifically for Republicans and Democrats. Test statistics are reported in Table S18-S21. Predictions by the developers of the treatments are reported in Tables S57-S66.

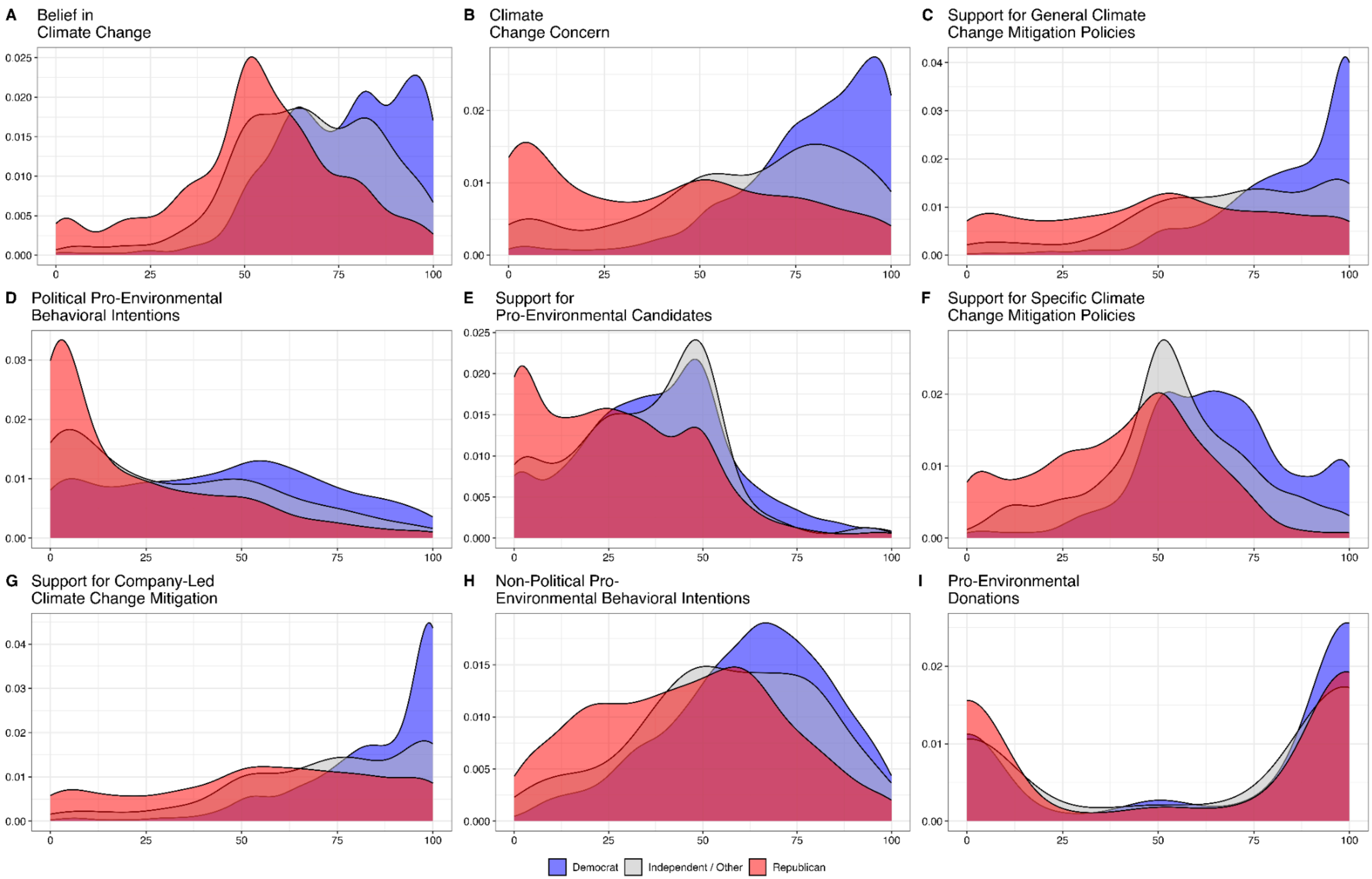

Fig. 1. Distributions of participants' pro-environmental attitudes, behavioral intentions, and behaviors. All outcomes range from 0 to 100. Ranges on the y axes differ for each variable. Partisan differences are shown by differences in distributions between Democrats (blue), Independents/Others (grey), and Republicans (red). Overlap between Democrats and Independents/Others is shown in blue-grey. Overlap between Independents/Others and Republicans is shown in grey-red. Overlap between all three groups is shown blue-grey-red.

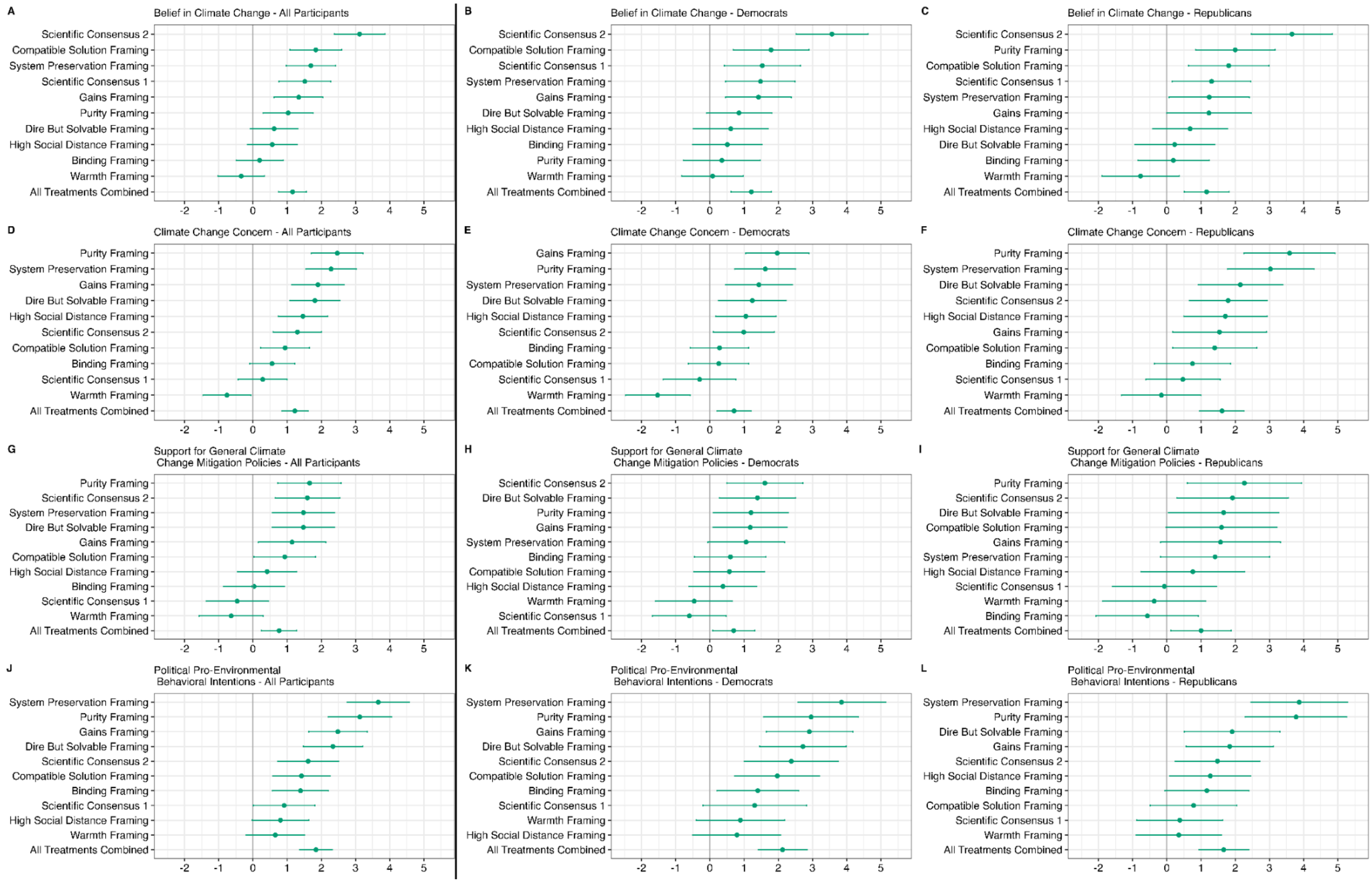

Fig. 2. Treatment effects on primary outcomes. All panels show unstandardized regression coefficients and 95% confidence intervals for the effects of the 10 treatments and the combination of all treatments, relative to the control condition, based on preregistered regression models (tables S18 to S21). All outcomes range from 0 to 100. Treatments are sorted in order of effect size. The left, middle, and right column reports the effects for the entire sample, Democrats, and Republicans, respectively.

Increasing Belief in Climate Change

Six of the ten treatments significantly increased belief in climate change, and none reduced belief in climate change. Overall, the treatment conditions increased belief in climate change by 1.16 pp ($p < .001$) compared to the control condition. To contextualize this magnitude of treatment effect, it represents 5% of the "partisan gap" at baseline, i.e., the difference between levels of climate change belief reported by Democrats compared to Republicans on the pre-treatment, baseline survey. The *Scientific Consensus 2* treatment had the largest effect (3.12 pp, $p < .001$, 14% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (five significant effects; no backfire effects; overall treatment effect: 1.17 pp, $p < .001$; largest effect for *Scientific Consensus 2*: 3.66 pp, $p < .001$) or Democrats (five significant effects; no backfire effects; overall effect: 1.21 pp, $p < .001$; largest effect for *Scientific Consensus 2*: 3.57 pp, $p < .001$).

Increasing Climate Change Concern

Seven treatments significantly increased climate change concern, and one treatment had a significant backfire effect. Overall, the treatment conditions increased climate change concern by 1.23 pp ($p < .001$, 3% of the partisan gap) compared to the control condition. The *Purity Framing* treatment had the largest effect (2.47 pp, $p < .001$, 7% of the partisan gap). *Warmth Framing* exhibited the backfire effect (-0.76 pp, $p = .034$, 2% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (seven significant effects; no backfire effects; overall effect: 1.61 pp, $p < .001$; largest effect for *Purity Framing*: 3.59 pp, $p < .001$) or Democrats (six significant effects; one backfire effect; overall effect: 0.71 pp, $p = .006$; largest effect for *Gains Framing*: 1.97 pp, $p < .001$; backfire effect for *Warmth Framing*: -1.53 pp, $p = .002$).

Increasing Support for General Climate Change Mitigation Policies

Six treatments significantly increased support for general climate change mitigation policies and none had a significant backfire effect. Overall, the treatment conditions increased support for general climate change mitigation policies by 0.76 pp (p = .003, 2% of the partisan gap) compared to the control condition. Again, the *Purity Framing* treatment had the largest effect (1.65 pp, p < .001, 5% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (three significant effects; no backfire effects; overall effect: 1.00 pp, p = .026; largest effect for *Purity Framing*: 2.27 pp, p = .008) or Democrats (four significant effects; no backfire effects; overall effect: 0.69 pp, p = .027; largest effect for *Scientific Consensus 2*: 1.61 pp, p = .005), although fewer treatment effects were significant.

Increasing Political Pro-Environmental Behavioral Intentions

Eight treatments significantly increased political pro-environmental behavioral intentions, and none had a significant backfire effect. Overall, the treatment conditions increased political pro-environmental behavioral intentions by 1.84 pp (p < .001, 8% of the partisan gap) compared to the control condition. Here, the *System Preservation Framing* treatment had the largest effect (3.67 pp, p < .001, 15% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (six significant effects; no backfire effects; overall effect: 1.66 pp, p < .001; largest effect for *System Preservation Framing*: 3.87 pp, p < .001) and Democrats (seven significant effects; no backfire effects; overall effect: 2.12 pp, p < .001; largest effect for *System Preservation Framing*: 3.85 pp, p < .001), although fewer treatment effects were significant.

Exploratory Analyses

Additional Outcomes

Figure 3 visualizes the treatment effects on the additional outcomes for the entire sample, Republicans and Democrats (using the same analysis strategy as for the preregistered outcomes

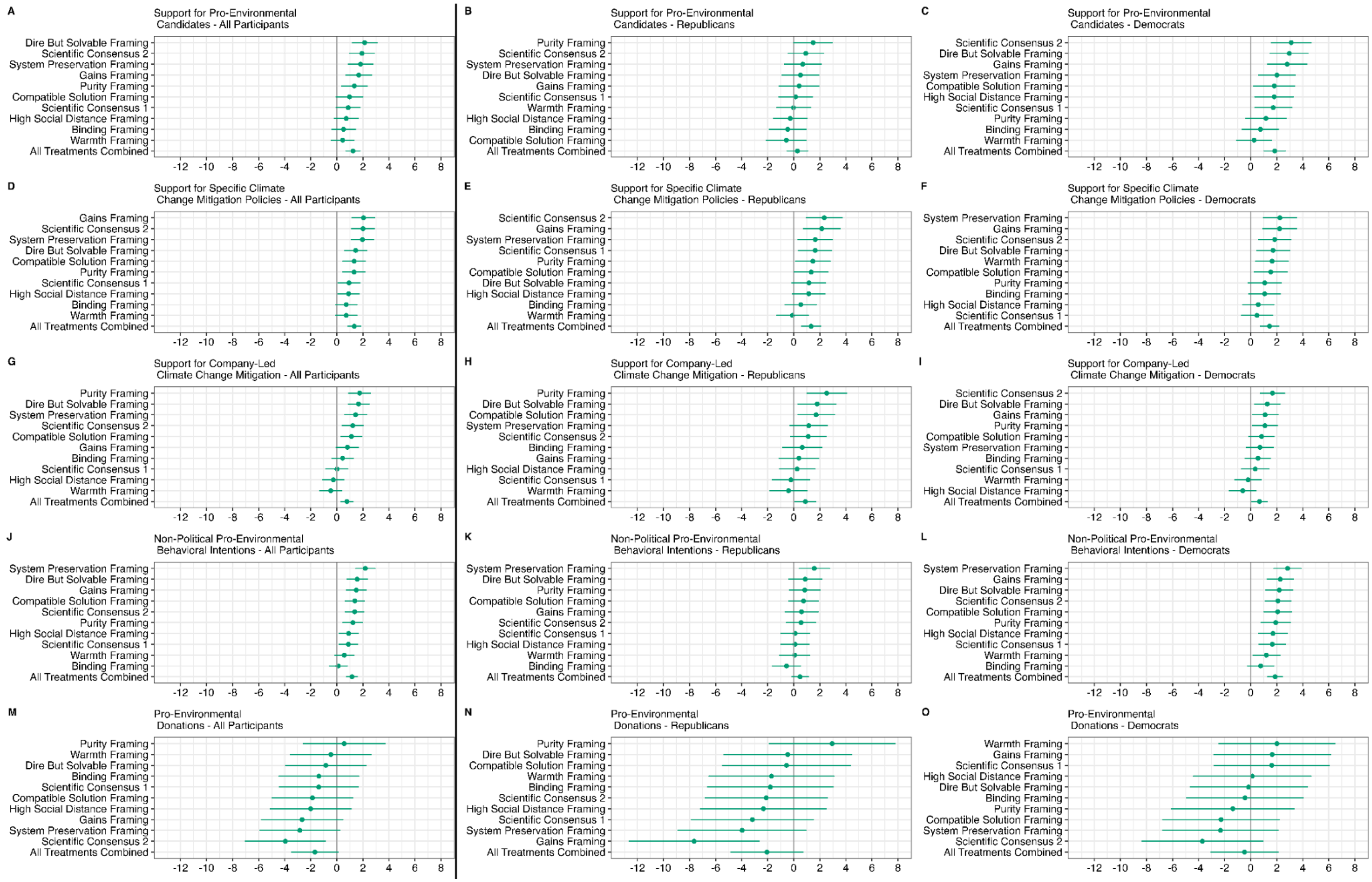

Fig. 3. Treatment effects on additional outcomes. All panels show unstandardized regression coefficients and 95% confidence intervals for the effects of the 10 treatments and the combination of all treatments, relative to the control condition, based on regression models parallel to the preregistered models reported in Fig. 2 (tables S22 to S26). All outcomes range from 0 to 100. Treatments are sorted in order of effect size. The left, middle, and right column reports the effects for the entire sample, Democrats, and Republicans, respectively.

except for the pro-environmental donations measure for which we controlled for pre-treatment measures of the other eight outcomes because we did not have a pre-treatment measure of pro-environmental donations). Test statistics are reported in Table S22-S26. Predictions by the developers of the treatments are reported in Tables S57-S66.

Increasing Support for Pro-Environmental Candidates

Five treatments significantly increased support for pro-environmental candidates. None of the treatments had a significant backfire effect. Overall, the treatment conditions increased support for pro-environmental candidates by 1.25 pp ($p < .001$, 11% of the partisan gap) compared to the control condition. The *Dire But Solvable Framing* treatment had the largest effect (2.14 pp, $p < .001$, 19% of the partisan gap). Smaller and fewer significant effects were obtained when restricting the sample to Republicans (one significant effect; no backfire effects; overall effect: 0.28 pp; $p = .496$, largest effect for *Purity Framing*: 1.49 pp, $p = .048$), whereas even more effects were obtained among Democrats (seven significant effects; no backfire effects; overall effect: 1.84 pp, $p < .001$; largest effect for *Scientific Consensus 2*: 3.11 pp, $p < .001$).

Increasing Support for Specific Climate Change Mitigation Policies

Eight treatments significantly increased support for specific climate change mitigation policies. None of the treatments had a significant backfire effect. Overall, the treatment conditions increased support for specific climate change mitigation policies by 1.34 pp ($p < .001$, 5% of the partisan gap) compared to the control condition. *Gains Framing* had the largest effect (2.05 pp, $p < .001$, 8% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (six significant effects; no backfire effects; overall effect: 1.33 pp, $p < .001$; largest effect for *Scientific Consensus 2*: 2.34 pp, $p = .001$) and Democrats (six significant effects; no backfire effects; overall effect: 1.44 pp, $p < .001$; largest effect for *System Preservation Framing*: 2.24 pp, $p = .001$).

Increasing Support for Company-Led Climate Change Mitigation

Five treatments significantly increased support for company-led climate change mitigation, and none had a significant backfire effect. Overall, the treatment conditions increased support for company-led climate change mitigation by 0.78 pp (p = .001, 3% of the partisan gap) compared to the control condition. *Purity Framing* had the largest effect (1.75 pp, p < .001, 6% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (three significant effects; no backfire effects; overall effect: 0.89 pp, p = .030; largest effect for *Purity Framing 2*: 2.53 pp, p = .001) and Democrats (four significant effects; no backfire effects; overall effect: 0.67 pp, p = .023; largest effect for *Scientific Consensus 2*: 1.67 pp, p < .001).

Increasing Non-Political Pro-Environmental Behavioral Intentions

Eight treatments significantly increased non-political pro-environmental behavioral intentions. None of the treatments had a significant backfire effect. Overall, the treatment conditions increased non-political pro-environmental behavioral intentions by 1.17 pp (p < .001, 7% of the partisan gap) compared to the control condition. *System Preservation Framing* had the largest effect (2.19 pp, p < .001, 14% of the partisan gap). Fewer effects were obtained when restricting the sample to Republicans (one significant effect; no backfire effects; overall effect: 0.48 pp, p = .138; largest effect for *System Preservation Framing*: 1.58 pp, p = .008), whereas even more effects were obtained among Democrats (nine significant effects; no backfire effects; overall effect: 1.87 pp, p < .001; largest effect for *System Preservation Framing*: 2.84 pp, p < .001).

Increasing Pro-Environmental Donations

No treatment significantly increased pro-environmental donations, though one of the treatments had a significant backfire effect. Overall, the treatment conditions did not have a significant effect on pro-environmental donations (-1.69 pp, p = .061, 14 % of the partisan gap) compared to the control condition. *Scientific Consensus 2* had the backfire effect (-3.95 pp, p = .012, 32% of the partisan gap). Similar results were obtained when restricting the sample to Republicans (no significant effect; one significant backfire effect; overall effect: -2.06 pp, p = .142; backfire effect for *Gains Framing*: -7.65 pp, p = .003), whereas no significant effects in either direction were obtained among Democrats (overall effect: -0.48 pp, p = .714).

Relative Efficacy of Treatments

In this section, we compare the efficacy of different treatments. We were particularly interested in top-performing treatments, i.e., whether the treatment with the largest effect size for a given outcome was statistically distinguishable from the other treatments. For most outcomes, there was not a clear, top-performing treatment. The exception was *Scientific Consensus 2*, which had a significantly stronger effect than all other treatments on belief in climate change among the full sample, and specifically among both Republicans and Democrats. For none of the other outcomes did the treatment with the largest effect size have significantly stronger effects than all other treatments. Detailed results are provided in Tables S27 - S35 in the SI.

Heterogeneous Treatment Effects

In this section, we explore whether treatment effects were moderated by variables that have been theorized to be important in the climate change communication literature. Although eight of the ten original papers found stronger effects among Republicans (or groups with associated beliefs such as conservatives, people with a strong belief in a just world, high levels of system justification, or strong free market support), there was a surprising similarity in treatment effects for Democrats and Republicans. First, treatments that had stronger effects on Democrats tended to have stronger effects on Republicans as well. Figure 4 shows the correlation of r = .40 between the 90 treatment effects (10 treatments x 9 outcomes) among Democrats (x-axis) and Republicans (y-axis). If we restrict the effects to the eight treatments for which the results of the

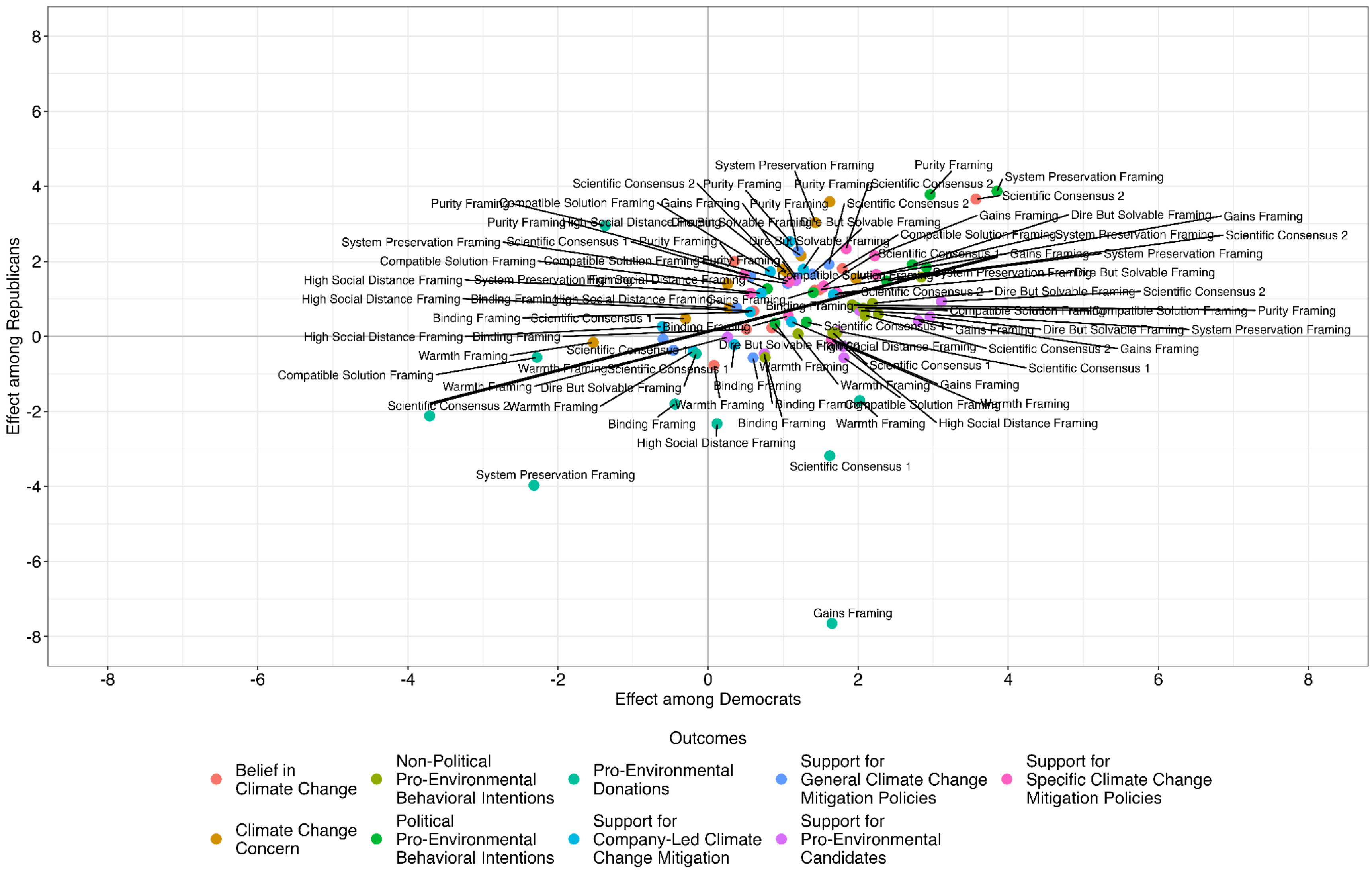

Fig. 4. Correlation of treatment effects among Democrats and Republicans. The x-axis (y-axis) provides the unstandardized regression coefficients for the effects of the 10 treatments on the nine outcomes, relative to the control condition, among Democrats (Republicans). The color of the points represent the outcome. The label of the points represents the treatment. All outcomes range from 0 to 100.

original papers would suggest stronger effects among Republicans, the correlation is even higher (r = .52). Second, treatment effects were also similar in size among Democrats and Republicans. Only 12 of the 90 treatment x partisan identity (Republican vs Democrat) interaction effects were significant (Figure 5). If we restrict the effects to the eight treatments for which the results of the original papers would suggest stronger effects among Republicans, 9 interactions effects were significant. However, for seven of the interaction effects, the pattern was the opposite: stronger effects among Democrats. The only treatment for which we found two significant interaction effects indicating stronger treatment effects among Republicans was *Purity Framing*.

Beyond partisan identity, we also explored other potential moderators, including strength of partisan identity, ideology, ideological extremity, belief in climate change, and climate change concern. Similarly, we found that treatment effects were correlated and significant interaction effects were rare for comparisons between non-partisans and partisans (r = .80; two significant interaction effects; Figures S1 and S2), liberal and conservative participants (r = .21; six significant interaction effects; Figures S3 and S4), people with moderate and strong ideologies (r = .72; seven significant interaction effects; Figures S5 and S6), those who believed less and more in climate change before the treatment (r = .56; ten significant interaction effects; Figures S7 and S8). The only exception was climate change concern. While treatment effects were strongly correlated among those who were less versus more concerned about climate change before the treatment (r = .52; Figure S9), we observed 26 significant interaction effects, of which 19 indicated that treatment effects were stronger when participants were more concerned about climate change (Figure S10).

Relationships between Outcomes

The test of many treatments in parallel allows us to examine to what extent pro-environmental attitudes, behavioral intentions, and behaviors were affected similarly by the

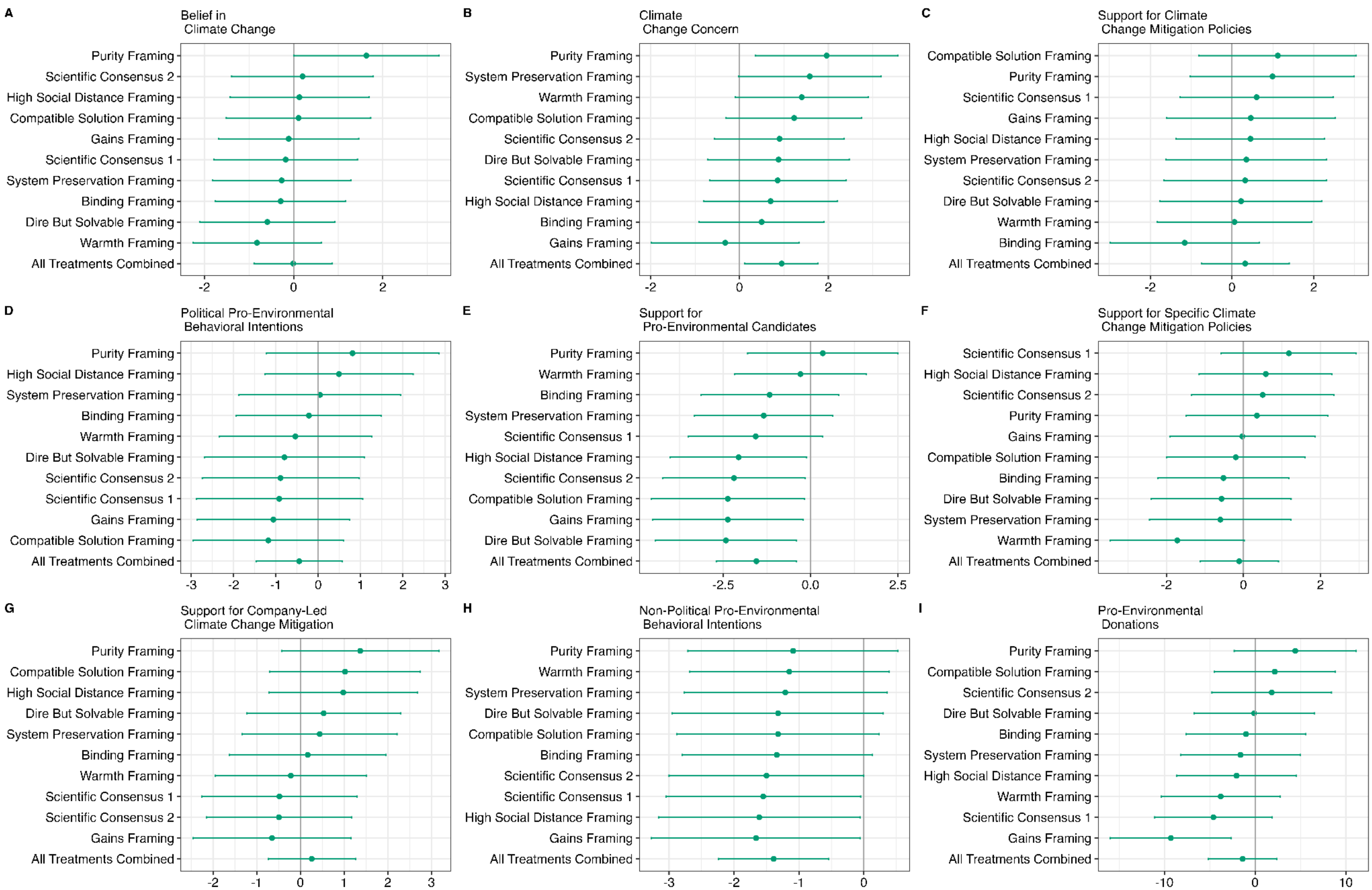

Fig. 5. Treatment x partisan affiliation interaction effects. All panels show unstandardized regression coefficients and 95% confidence intervals for treatment (ref: control condition) x partisan affiliation (Republican vs Democrat) interaction effects for the 10 treatments and the combination of all treatments. All outcomes range from 0 to 100. Treatments are sorted in order of effect size. Positive interaction effects indicate stronger treatment effects among Republicans.

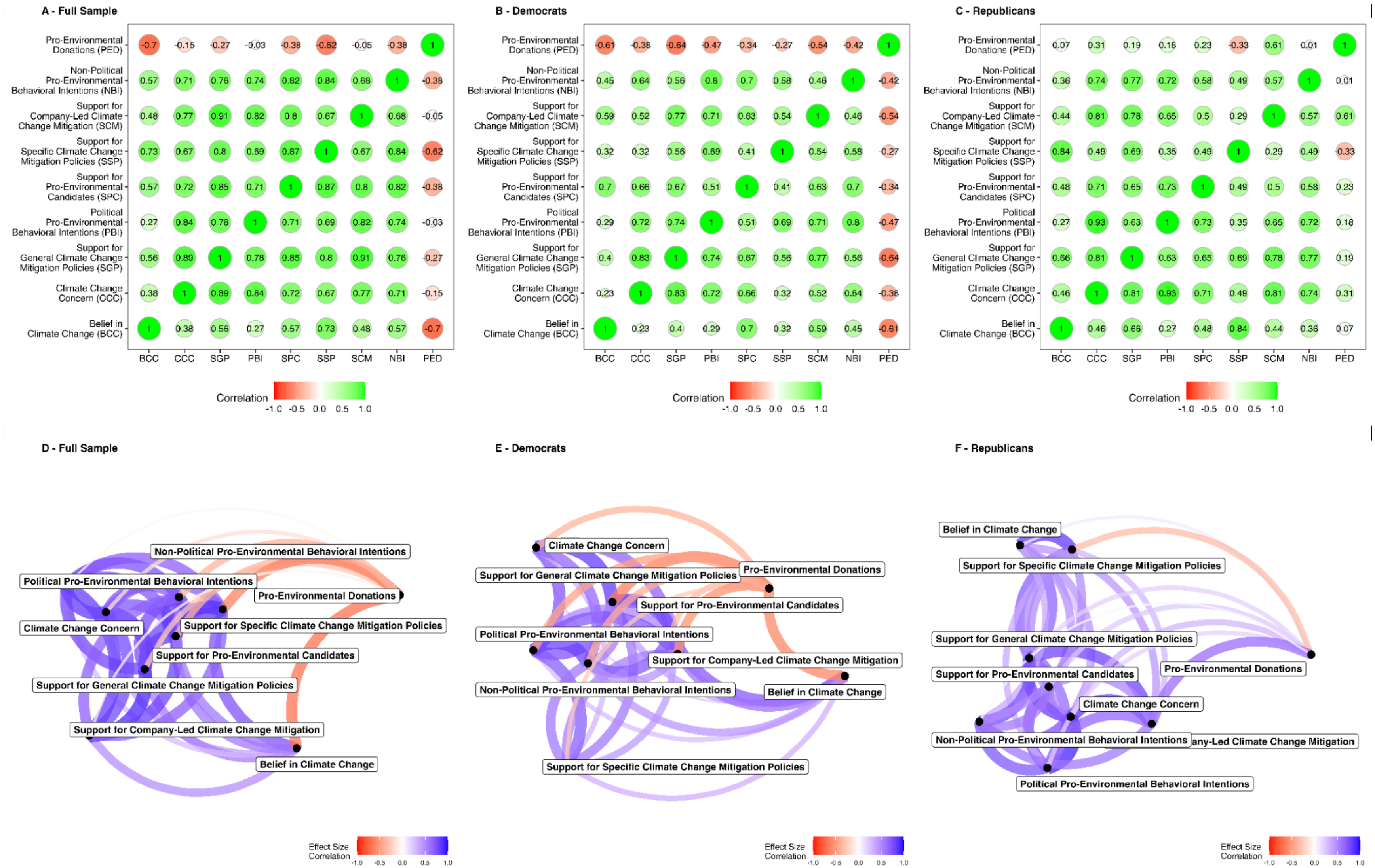

Fig. 6. Relationships between outcomes. Panels A-C show (A) network diagrams visualizing Pearson correlations calculated using Cohen's ds across the 10 treatments, for each pair of outcome variables. Classical multidimensional scaling was used to calculate two-dimensional coordinates for each outcome. Closer outcomes indicate closer similarities, i.e. positively correlated treatment effects. Panels D-F show the correlation matrices with the pairwise correlation coefficients, visualized in Panels A-C. The left, middle, and right column reports the effects for the entire sample, Democrats, and Republicans, respectively.

same treatments. Figure 6A shows pairwise correlations of the ten treatments' effects on the nine outcomes. Figure 6D visualizes these correlations in a network diagram in which outcome variables that were affected by treatments similarly are located more closely together and linked by a stronger tie.

We find that treatment effects on all eight pro-environmental attitudes and behavioral intentions were always positively and often strongly correlated (range of r's$_{\text{effect size}}$ = [.27, .91]). In contrast, treatment effects on donations to pro-environmental organizations were *negatively* correlated with effects on the other eight outcomes (range of r's$_{\text{effect size}}$ = [-.70, -.03]). Notably, these negative correlations were driven by Democrats (Figures 6B and 6E), whereas treatment effects on donations to pro-environmental organizations tended to be positively correlated with effects on the other eight outcomes among Republicans (Figures 6C and 6F).

These findings suggest that the various pro-environmental attitudes and behavioral intentions we studied are closely linked, responding in similar ways to different experimental treatments with quite variable content. However, we also find that treatments that causally influence these pro-environmental attitudes and behavioral intentions do not necessarily have similar effects on pro-environmental behaviors. Indeed, we found that the same causes that led attitudes and behavioral intentions to become more pro-environment tended to have no influence on, or even slightly reduce, pro-environmental donations, a pattern that suggests pro-environmental attitudes and intentions share common antecedents, whereas pro-environmental behaviors like donating may be influenced by distinct factors.

Mechanisms Driving Treatment Effects

The messages that we tested can be better thought of as manipulating multiple constructs, rather than a single construct, raising doubts about clean inferences about theoretical mechanisms. We tested the treatment effects on nine manipulation check items - one for each construct described as the primary mechanism underlying each of the treatment messages (note that we had two scientific consensus messages). We found that none of the ten treatments had significant effects on just the one manipulation check corresponding to their intended construct (see Figure 7). Two treatments - *High Social Distance Framing* and *Warmth Framing* - did not affect any of the manipulation checks, suggesting that these messages were not good tests of the effects of the construct they sought to manipulate. The other eight treatments affected two to five manipulation checks, suggesting that these messages treated multiple theoretical constructs. For example, *Purity Framing* did not significantly increase the manipulation check corresponding to purity–the perceived disgustingness of climate change. However, it increased how patriotic climate change mitigation is, how much climate change mitigation protects the American way of life, and the personal relevance of climate change mitigation.

We next explore potential causal mechanisms driving observed treatment effects by estimating correlations between (a) effects of the treatments on the manipulation checks, and (b) effects of the treatments on the primary outcome variables. This analysis offers insights on how similarly the treatments affected the manipulation check items and the primary outcome variables. In the full sample (see Figure 8A), we find that treatment effects on a composite of the four primary outcome variables were most correlated with treatment effects on the following manipulation checks: the perceived patriotism of climate change mitigation ($r = .73$), the perceived impact of climate change mitigation on the American way of life ($r = .72$), disgust felt toward environmental pollution ($r = .61$), and the perceived personal relevance of climate change mitigation ($r =.54$). Among Democrats (see Figure 8B), treatment effects on these four manipulation checks were even more strongly associated with treatment effects on the composite (r'a range from .59 to .82). Among Republicans (see Figure 8C), the effect size correlations were

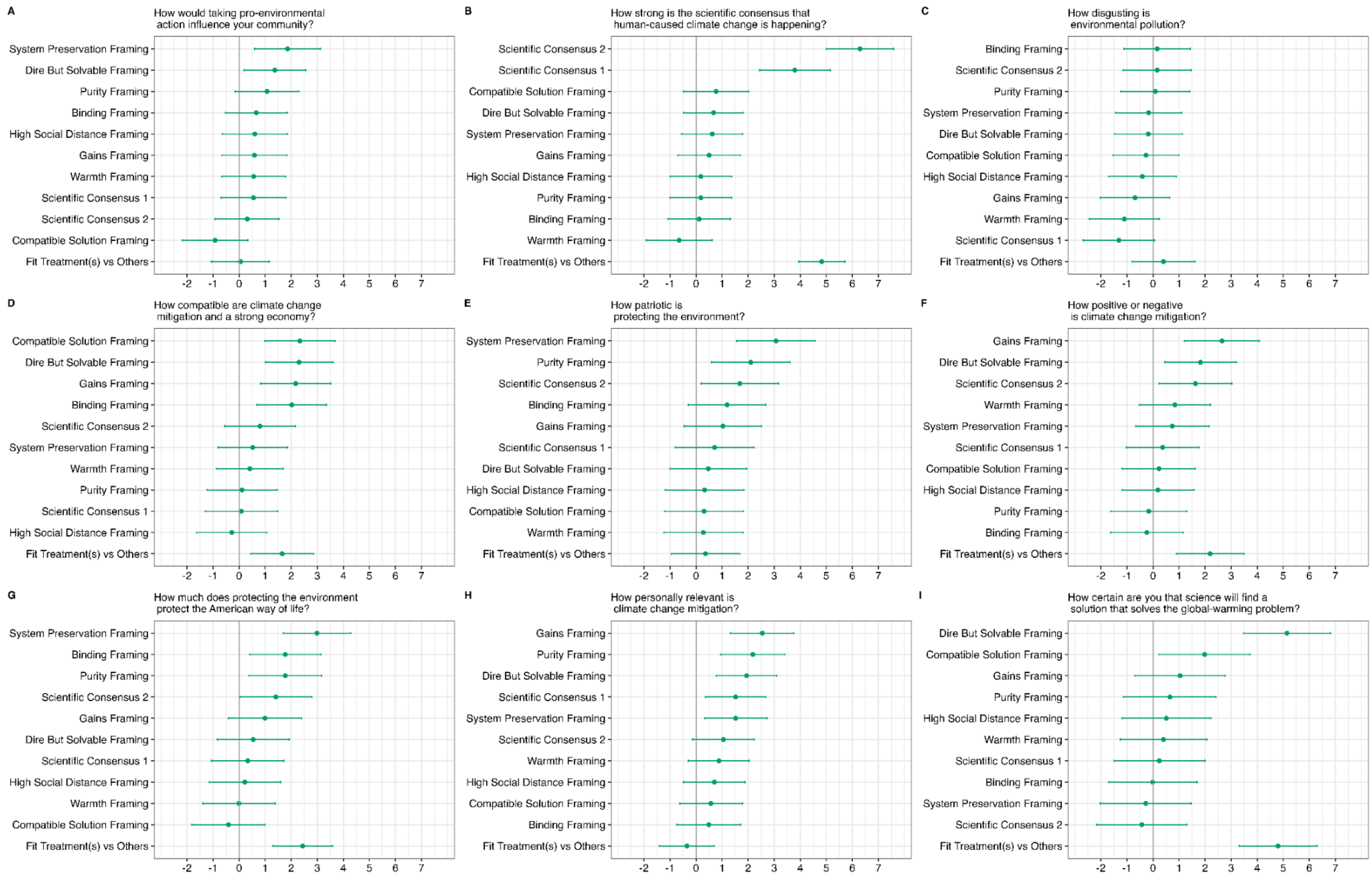

Fig. 7. Treatment effects on manipulation checks. All panels show unstandardized regression coefficients and 95% confidence intervals for the effects of the 10 treatments, relative to the control condition, and the combination of treatments designed to move the manipulation, relative to all other conditions, based on regression models parallel to the preregistered models reported in Fig. 2. All outcomes range from 0 to 100. Treatments are sorted in order of effect size.

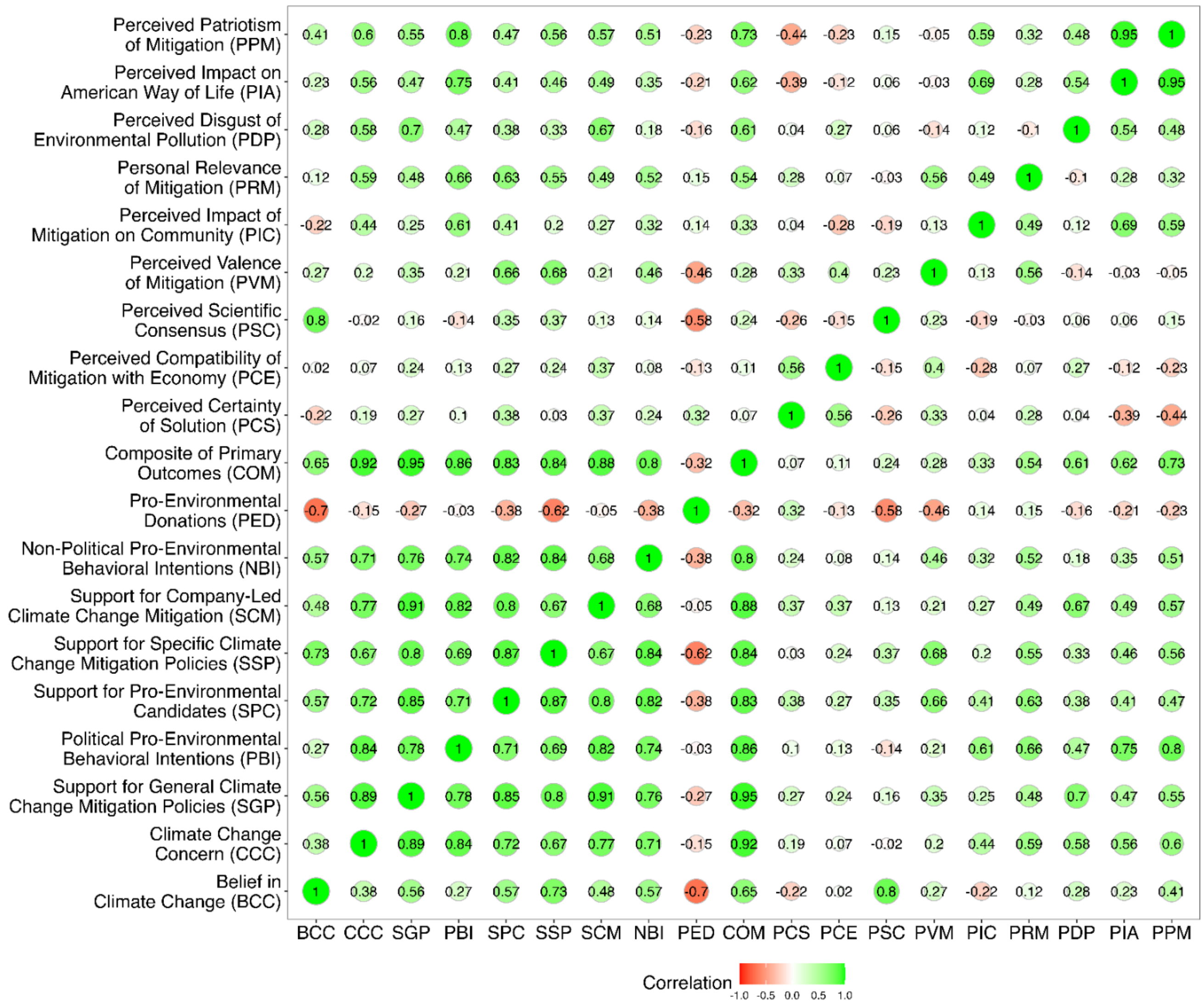

Fig. 8. Relationships between outcomes and manipulation checks. The correlation matrix shows Pearson correlations calculated using Cohen's d's across the 10 treatments, for each pair of outcome and manipulation check variables.

much weaker (r's range from -.06 to .43). These findings suggest multiple potential mechanisms likely drove the observed treatment effects, and effects on measures of these mechanisms were especially associated with effects on the outcome variables for Democrats.

Discussion

Our study advances the science of climate change communication in several ways. Several failed replication studies (see Pilot Data) motivated us to test the effects of ten widely cited treatments for increasing pro-environmental attitudes and behavioral intentions. Conducting a megastudy (Duckworth & Milkman, 2023; Voelkel et al., 2024b) - experiments in which all messages were tested in parallel with the same sample, a shared control condition, and common dependent measures - allowed us to also estimate not only the absolute treatment effects, but also the relative efficacy of these treatments.

We find that many of the ten treatments we tested strengthened the pro-environmental attitudes and behavioral intentions of Americans in general, as well as those of both Democrats and Republicans specifically. Not every single treatment significantly affected every outcome variable, though nine of the ten treatments affected at least one of the four primary outcomes in the study. For each of our primary outcomes, more than five treatments had significant positive effects. At the same time, we found little evidence for specific messages performing significantly better than all others for a given outcome variable, with the exception of *Scientific Consensus 2,* which had a significantly stronger effect on belief in climate change (though not any other outcomes) than all other treatments.

When message treatments did have significant effects, they did not cause major shifts in Americans' views, with the size of the largest treatment effects corresponding to just 5% to 19% of the gap between the reported views of Democrats and Republicans on each outcome.

Additionally, there is good reason to expect the size of these effects would diminish over time (e.g., Voelkel et al., 2024). Thus, for climate change messaging to achieve substantial and durable effects, it likely would be necessary for messages to be administered repeatedly, or in larger doses, ideally by structurally embedding efficacious messages in individuals' information environments, for example, through communication by political elites, modification of social media feeds, the content of curricula in schools, or widely coordinated advertising campaigns.

Why did our megastudy show more evidence for the effectiveness of widely cited messaging strategies than the pilot studies? One reason is that the megastudy was far better powered to detect small effect sizes. The increase in power was due to an increase in sample size and our use of a pre-post design. In addition to the boost in power, the pre-post design offered valuable pre-treatment measures of potential moderators, helping us assess potential biases on causal inference from attrition. Recent research (Clifford et al., 2021) suggests that long-standing concerns that repeated measures designs like this may suppress treatment effects because participants become more committed to their initial attitudes, and more likely to subsequently respond consistently with them, are exaggerated. While we cannot directly test the effects of the repeated measures design might have had on estimated effect sizes, our effect sizes for belief in climate change and support for general climate change mitigation policies were comparable to estimated effects in another megastudy that used similar treatments and measures of these two constructs (Vlasceanu et al., 2024). These findings suggest our use of a pre-post design was unlikely to have invoked potential consistency pressures that reduced estimated effects sizes.

Another possible explanation is that the pilot studies we conducted involved mostly direct replications using original stimuli verbatim, while in the megastudy the original authors were allowed to revise their treatments, and eight of ten author teams did in fact revise their treatments. It is possible that the authors' revisions improved the efficacy of the treatments, at least their efficacy at the time the megastudy was conducted. It is unclear, however, if any improved efficacy was a result of the treatments being better tailored to the time and population in which the replication was conducted.

Our results do not offer clear implications regarding what general mechanisms shape Americans' climate change-related attitudes. Eight of the ten treatments had significant effects on multiple manipulation check items. The other two treatments - High Social Distance Framing and Warmth Framing - did not significantly affect any of these items. The fact that most treatments influenced multiple measures of potential mechanisms complicates inference about what is driving the observed effects. More generally, our design does not allow strong inference about causal relationships between the measured mechanisms and the outcomes due to the limitations of correlational mediation analysis (e.g., Bullock et al., 2012).

Assuming that the assumptions of mediation analyses hold, separate causal mediation analyses suggest that multiple mediators could drive several of the treatments with the largest effects. For example, 8%, 7%, 6%, and 5% of the effect of System Preservation Framing on a composite of the four primary outcome variables was mediated by participants' views that climate change mitigation protects the American way of life (the expected mediator), positively impacts one's community, is patriotic, and is personally relevant, respectively. Similarly, 10%, 5%, and 5% of the effect of Purity Framing on the composite was mediated by participants' views that climate change mitigation is personally relevant, protects the American way of life (the expected mediator), and is patriotic, respectively. It is noteworthy that the expected mediator, participants' views of how disgusting environmental pollution is, mediated less than 1% of the Purity Framing treatment effect. 26%, 4%, 4%, and 4% of the effect of Scientific

Consensus 2 on the composite of the four primary outcomes was mediated by participants' perceptions that a scientific consensus exists on human-caused climate change (the expected mediator), that climate mitigation protects the American way of life, and that climate mitigation policies are positive and patriotic, respectively. An important limitation of these analyses is that measurement error may impact the estimates because the manipulation check items were measured with single items. More research is needed that cleanly and independently manipulates the potential drivers of climate change attitudes. Our findings indicate that multiple mechanisms contribute to the effects of the climate change messaging treatments we tested, but the complexity of these influences and potential measurement error limit clear causal inference.

Our effect size correlational analysis suggests that perceived patriotism and personal relevance of climate change mitigation, the perceived impact of climate change on the American way of life, and feelings of disgust about environmental pollution ($r = .61$) were the most influential mechanisms in this study and, interestingly, were more associated with effect sizes among Democrats than among Republicans, even though moral foundations theory and system justification theory would suggest the opposite. Future research could build on these findings by testing (a) the effectiveness of highly controlled treatments carefully designed to employ each of these specific causal mechanisms, and no other mechanisms, and (b) the impact of a multifactorial treatment combining multiple of these mechanisms, to potentially increase overall effect size.

Our results also call into question the widespread belief that different strategies are needed to strengthen the pro-environmental attitudes and behavioral intentions of Republicans and Democrats (or worldviews associated with partisanship). Many of the treatments (*Binding Framing* (Wolsko et al., 2016); *Dire But Solvable Framing* (Feinberg & Willer, 2011);

Compatible Solution Framing (Campbell & Kay, 2014); *High Social Distance Framing* (Hart & Nisbet, 2012); *Purity Framing* (Feinberg & Willer, 2013); *System Preservation Framing* (Feygina et al., 2010); *Warmth Framing* (Bain et al., 2012)) we tested were reported to be more efficacious among people who lean Republican or having worldviews associated with Republicans. Some of these treatments were indeed efficacious among Republicans but similarly changed attitudes among Democrats. The majority of the few significant interactions suggested that more persuasive gains can be made by increasing levels of support for pro-environmental candidates among Democrats instead of Republicans. While the megastudy was underpowered to rule out the existence of small interaction effects, our findings nonetheless offer an important corrective for a literature that often claims different messaging strategies are needed for Democrats and Republicans. Together with other recent results (Voelkel et al., 2022), our findings suggest that messages rooted in values that are traditionally associated with Republicans may also persuade with Democrats and Independents (but see also Feinberg & Willer, 2019).

More generally, we found little evidence for heterogeneous treatment effects. The only exception is that the effects of treatments often depended on participants' climate change concern. Interestingly, we found that people who were *more* concerned about climate change pre-treatment tended to show *larger* treatment effects, a notable pattern given that many treatment messages were specifically designed to target groups (e.g., Republicans, conservatives) who report less concern about climate change. Our results extend previous work finding little heterogeneity in treatment effects with demographic moderators (Coppock, 2023).

What do we learn about the relationship between pro-environmental attitudes, behavioral intentions, and behaviors? Although our sample size at the treatment level is small (n = 10), our results suggest that belief in climate change, climate change concern, support for climate change mitigation policies (both general and specific), support for company-led climate change mitigation, as well as political and non-political pro-environmental behavioral intentions are closely related. By implication, treatments that strengthen one of these outcomes tend to strengthen many of them. This pattern held for both Democrats and Republicans.

In contrast, treatment effects on the particular pro-environmental behaviors we studied - donations to pro-environmental organizations - were uncorrelated or even negatively associated with treatment effects on pro-environmental attitudes and behavioral intentions. This was particularly true among Democrats. This finding suggests that moving attitudes and behavioral intentions alone is not enough to change at least this behavior. Although our behavioral dependent variable was not preregistered, we conclude that while communication strategies can change people's attitudes, downstream effects on behaviors may be more difficult to achieve. Based on recent research suggesting that treatments are most efficacious when tailored to fit the context and goals of a specific application (Dechezleprêtre et al., 2025; Voelkel at al., 2024), efforts to increase pro-environmental behaviors should target behaviors directly, rather than inducing attitude change.

Our main conclusion from this research is that several well-cited messaging strategies reliably influence pro-environmental attitudes and behavioral intentions. However, the effect sizes are small, suggesting that achieving lasting impact may require messaging campaigns that involve repeated or uniquely impactful messaging. We found little evidence of heterogeneous treatment effects between Democrats and Republicans in the studied interventions, findings that are inconsistent with the theoretical bases of several of the tested messages. Future research is needed to identify the causal mechanisms that drive effective climate change messaging strategies.

Acknowledgement

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Competing Interests

One member of our core author team (RW) co-authored two of the ten messaging strategies (Dire but Solvable Framing and Purity Framing) tested in this experiment. We addressed the potential for bias resulting from a possible conflict of interest in several ways. First, we conducted this study based on failed replication studies, including studies testing Dire but Solvable Framing and Purity Framing messages. Therefore, RW expected that a failure to replicate the original findings was likely. Though we ended up finding several significant main effects, our experimental results failed to replicate some of the findings of the prior research conducted by RW and colleagues. This highlights our commitment to publishing results independent of their alignment with prior research. Second, data handling and analysis was conducted by a member of our research team who was not involved in the original papers. Third, the questionnaire and analysis script was preregistered so that researcher degrees of freedom was minimized. Fourth, RW recused himself from the process of reviewing the acceptability of changes to the text of the messages made by his team or any of the other original research teams.

Table 1: *Design Table*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies significantly increase belief in climate change?	For each messaging strategy: The treatment will significantly increase belief in climate change compared to the control condition.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model: $\text{Belief}_{i,\ 2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Belief}_{i,\ t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increased belief in climate change.
Which messaging strategies significantly increase climate change concern?	For each messaging strategy: The treatment will significantly increase climate change concern compared to the control condition.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model: $\text{Concern}_{i,\ t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Concern}_{i,\ t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increased climate change concern.
Which messaging strategies significantly increase support for general climate change mitigation policies?	For each messaging strategy: The treatment will significantly increase support for general climate change mitigation policies compared to the control condition.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model: $\text{Policy Support}_{i,\ t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Policy Support}_{i,\ t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increased support for general climate change mitigation policies.
Which messaging strategies significantly increase political pro-environmental behavioral intentions?	For each messaging strategy: The treatment will significantly increase political pro-environmental behavioral intentions compared to the control condition.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model: $\text{Intentions}_{i,\ t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Intentions}_{i,\ t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increased political pro-environmental behavioral intentions.

Table 1: *Design Table (continued)*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies significantly increase belief in climate change among Democrats?	For each messaging strategy: The treatment will significantly increase belief in climate change compared to the control condition among Democrats	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model: $Belief_{i,\ t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Belief_{i,\ t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases belief in climate change among Democrats.
Which messaging strategies significantly increase climate change concern among Democrats?	For each messaging strategy: The treatment will significantly increase climate change concern compared to the control condition among Democrats.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model: $Concern_{i,\ t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Concern_{i,\ t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases climate change concern among Democrats.
Which messaging strategies significantly increase support for general climate change mitigation policies among Democrats?	For each messaging strategy: The treatment will significantly increase support for general climate change mitigation policies compared to the control condition among Democrats.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model: $Policy\ Support_{i,\ t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Policy\ Support_{i,\ t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases support for general climate change mitigation policies among Democrats.
Which messaging strategies significantly increase political pro-environmental behavioral intentions among Democrats?	For each messaging strategy: The treatment will significantly increase political pro-environmental behavioral intentions compared to the control condition among Democrats.	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model: $Intentions_{i,\ t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Intentions_{i,\ t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases political pro-environmental behavioral intentions among Democrats.

Table 1: *Design Table (continued)*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies significantly increase belief in climate change among Republicans?	For each messaging strategy: The treatment will significantly increase belief in climate change compared to the control condition among Republicans.	98% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model: $\text{Belief}_{i,\,t=2}$ = b0 + b1 * Condition1_i + b2 * Condition2_i + … + b10 * Condition10i + $\text{Belief}_{i,\,t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases belief in climate change among Republicans.
Which messaging strategies significantly increase climate change concern among Republicans?	For each messaging strategy: The treatment will significantly increase climate change concern compared to the control condition among Republicans.	98% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model: $\text{Concern}_{i,\,t=2}$ = b0 + b1 * Condition1_i + b2 * Condition2_i + … + b10 * Condition10_i + $\text{Concern}_{i,\,t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases climate change concern among Republicans.
Which messaging strategies significantly increase support for general climate change mitigation policies among Republicans?	For each messaging strategy: The treatment will significantly increase support for general climate change mitigation policies compared to the control condition among Republicans.	98% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model: Policy $\text{Support}_{i,\,t=2}$ = b0 + b1 * Condition1_i + b2 * Condition2_i + … + b10 * Condition10_i + Policy $\text{Support}_{i,\,t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases support for general climate change mitigation policies among Republicans.
Which messaging strategies significantly increase political pro-environmental behavioral intentions among Republicans?	For each messaging strategy: The treatment will significantly increase political pro-environmental behavioral intentions compared to the control condition among Republicans.	98% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model: $\text{Intentions}_{i,\,t=2}$ = b0 + b1 * Condition1_i + b2 * Condition2_i + … + b10 * Condition10_i + $\text{Intentions}_{i,\,t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases political pro-environmental behavioral intentions among Republicans.

Table 1: *Design Table (continued)*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies most strongly increase belief in climate change?	Exploratory	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model:* $\text{Belief}_{i,\, t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Belief}_{i,\, t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases belief in climate change more strongly than the reference treatment.
Which messaging strategies most strongly increase climate change concern?	Exploratory	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model:* $\text{Concern}_{i,\, t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Concern}_{i,\, t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases climate change concern more strongly than the reference treatment.
Which messaging strategies most strongly increase support for general climate change mitigation policies?	Exploratory	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model:* $\text{Policy Support}_{i,\, t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Policy Support}_{i,\, t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases support for general climate change mitigation policies more strongly than the reference treatment.
Which messaging strategies most strongly increase political pro-environmental behavioral intentions?	Exploratory	99% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Everyone Method: OLS regression Model:* $\text{Intentions}_{i,\, t=2} = b0 + b1 * \text{Condition1}_i + b2 * \text{Condition2}_i + \ldots + b10 * \text{Condition10}_i + \text{Intentions}_{i,\, t=1}$	A positive and statistically significant effect ($p < .05$) for a treatment will be interpreted as evidence that this treatment increases political pro-environmental behavioral intentions more strongly than the reference treatment.

Table 1: *Design Table (continued)*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies most strongly increase belief in climate change among Democrats?	Exploratory	87% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model:* $Belief_{i,\,t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Belief_{i,\,t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases belief in climate change more strongly than the reference treatment among Democrats.
Which messaging strategies most strongly increase climate change concern among Democrats?	Exploratory	87% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model:* $Concern_{i,\,t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Concern_{i,\,t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases climate change concern more strongly than the reference treatment among Democrats.
Which messaging strategies most strongly increase support for general climate change mitigation policies among Democrats?	Exploratory	87% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model:* $Policy\ Support_{i,\,t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Policy\ Support_{i,\,t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases support for general climate change mitigation policies more strongly than the reference treatment among Democrats.
Which messaging strategies most strongly increase political pro-environmental behavioral intentions among Democrats?	Exploratory	87% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Democrats Method: OLS regression Model:* $Intentions_{i,\,t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Intentions_{i,\,t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases political pro-environmental behavioral intentions more strongly than the reference treatment among Democrats.

Table 1: *Design Table (continued)*

Research Question	Hypothesis	Sampling plan	Analysis Plan	Interpretation
Which messaging strategies most strongly increase belief in climate change among Republicans?	Exploratory	88% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model:* $Belief_{i, t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Belief_{i, t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases belief in climate change more strongly than the reference treatment among Republicans.
Which messaging strategies most strongly increase climate change concern among Republicans?	Exploratory	88% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model:* $Concern_{i, t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Concern_{i, t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases climate change concern more strongly than the reference treatment among Republicans.
Which messaging strategies most strongly increase support for general climate change mitigation policies among Republicans?	Exploratory	88% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model:* $Policy\ Support_{i, t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Policy\ Support_{i, t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases support for general climate change mitigation policies more strongly than the reference treatment among Republicans.
Which messaging strategies most strongly increase political pro-environmental behavioral intentions among Republicans?	Exploratory	88% power to detect effect sizes of Cohen's $d \geq 0.10$	Sample: Only Republicans Method: OLS regression Model:* $Intentions_{i, t=2} = b0 + b1 * Condition1_i + b2 * Condition2_i + \ldots + b10 * Condition10_i + Intentions_{i, t=1}$	A positive and statistically significant effect (p < .05) for a treatment will be interpreted as evidence that this treatment increases political pro-environmental behavioral intentions more strongly than the reference treatment among Republicans.

* Notes: Multiple models with different reference categories for Condition will be run.

Treatment	Description
Binding Framing	Participants read a text arguing that mitigating climate change is consistent with values such as patriotism, family, and religiosity.
Scientific Consensus I	Participants read a text emphasizing the scientific consensus that human-caused climate change is happening and inoculating readers against narratives undermining this consensus.
Scientific Consensus II	Participants read a text emphasizing the scientific consensus that human-caused climate change is happening and inoculating readers against narratives undermining this consensus.
Dire but Solvable Framing	Participants read a text arguing that the consequences of unaddressed climate change are dire but that there are solutions to address climate change.
Compatible Solution Framing	Participants read a text arguing that climate change can be mitigated in ways that are consistent with free market principles.
Gains Framing	Participants read a text highlighting what can be gained from mitigating climate change.
High Social Distance Framing	Participants read a text highlighting how a socially distant group for many Americans (Chinese farmers) is harmed by climate change.
Purity Framing	Participants read a text arguing that mitigating climate change is consistent with values such as protecting the purity of American land.
System Preservation Framing	Participants read a text arguing that mitigating climate change is consistent with values such as preserving the traditional American way of life.
Warmth Framing	Participants read a text arguing that mitigating climate change is consistent with values such as caring for fellow citizens.

Table 2 | The 10 treatments. The label column provides the names of the treatments. The description column provides a summary of the treatment. More information about the treatments (including the developer of the treatments) can be found in SI sections Original Treatments and Revised Treatments.

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